

Dynamic Graph Transformer-Convolutional Network for Multi-Channel EMG Gesture Recognition*

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Abstract—Electromyography (EMG) signal classification is a critical task in biomedical engineering, particularly for hand gesture recognition, which demands accurate modeling of the temporal and spatial dependencies within multi-channel EMG data. In this study, we propose a novel Dynamic Graph Transformer-Convolutional Network (GTCN) to address the unique challenges of EMG signal classification. The model leverages a dynamic graph convolutional network (GCN) to capture the inter-channel relationships dynamically at each time step and a Transformer architecture to model long-range temporal dependencies effectively. In the dynamic graph, each node represents the functional coordination of specific muscles during a gesture, while the dynamic graph structure reflects the strength of cooperative interactions between muscles as they vary throughout the gesture. This design enables the model to adaptively represent the evolving spatial dependencies among EMG channels during movement. The Transformer encoder further enhances the temporal modeling by attending to salient patterns in the sequential data, while the GCN ensures robust spatial feature extraction. We evaluate the proposed algorithm on a publicly available EMG dataset for hand gesture recognition. Experimental results demonstrate that the GTCN achieves an accuracy of 97.24%, outperforming several state-of-the-art methods. This performance validates the model's capability to effectively exploit the temporal and spatial characteristics inherent in EMG signals, particularly the functional synergy and varying cooperation among muscles during gestures, offering a promising approach for advanced EMG-based gesture recognition tasks.

I. INTRODUCTION

Electromyography (EMG) signal recognition has become a cornerstone in advancing technologies for human-machine interaction, rehabilitation, and prosthetic control. By analyzing the electrical activity generated by muscle contractions, EMG-based systems enable intuitive and precise control of external devices, such as robotic arms, exoskeletons, and virtual interfaces [1]. These systems hold immense potential for improving the quality of life for individuals with physical disabilities, enhancing productivity in industrial applications, and enabling immersive experiences in virtual and augmented reality [2]. The accurate recognition of EMG signals

for hand gestures, however, remains a significant challenge due to the complex, non-stationary, and multi-dimensional nature of the data [3], underscoring the need for innovative and robust computational methods.

Traditional methods for EMG gesture recognition typically rely on manual feature extraction combined with classical machine learning algorithms such as support vector machines (SVMs), random forests, and k-nearest neighbors (KNNs) [4]. Feature extraction techniques often include time-domain, frequency-domain, and statistical features to characterize EMG signals [5]. While these approaches have shown promise, they are heavily dependent on domain expertise and often fail to generalize across diverse subjects and recording conditions. Furthermore, their limited ability to capture the non-linear and multi-dimensional dependencies inherent in EMG data has posed a significant barrier to achieving high recognition accuracy in complex real-world scenarios.

In recent years, deep learning-based methods have gained popularity for EMG gesture recognition due to their ability to automatically extract high-level features from raw data. Techniques such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), and their hybrid forms have been extensively employed [6]. These methods have demonstrated improved performance by learning spatial and temporal dependencies within EMG signals. However, most deep learning models treat EMG channels independently, overlooking the cooperative interactions among muscles that occur during gesture execution [7]. Moreover, these models often require large-scale labeled datasets and lack interpretability in terms of the underlying biomechanical relationships.

EMG signals exhibit unique characteristics, including non-stationary temporal patterns and spatially distributed signals across multiple channels [8]. These signals reflect the coordinated activity of muscles that work synergistically to perform gestures. For instance, specific gestures involve dynamic changes in the functional connectivity and cooperation between muscle groups. Capturing these interactions is crucial for accurately modeling EMG data. The challenge lies in designing a framework that can adaptively model both the temporal evolution of EMG signals and the dynamic inter-channel relationships that represent muscle coordination.

To address these challenges, we propose a novel dynamic Graph Transformer-Convolutional Network (GTCN). The model integrates a dynamic graph convolutional network (GCN) to adaptively capture the evolving spatial relationships between EMG channels and a Transformer architecture to model long-range temporal dependencies. By representing

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each EMG channel as a graph node and iteratively updating the graph structure to reflect the cooperative strength between muscles during gestures, the DGT-CN effectively encodes the synergy of muscle groups. The Transformer component further enhances the model's ability to attend to salient temporal features, leading to robust gesture recognition. The main contributions of this study are summarized as follows:

1. We propose a novel GTCN model that combines dynamic GCN and Transformer to effectively exploit both the spatial and temporal characteristics of EMG signals.
2. We design a dynamic graph mechanism that reflects the evolving coordination and cooperation between muscles during gestures, offering enhanced interpretability of the recognition process.
3. We validate the proposed approach on a publicly available EMG dataset, demonstrating its superior performance, outperforming existing state-of-the-art methods.
4. Our model provides insights into the biomechanical aspects of gesture execution, bridging the gap between computational methods and muscle function analysis.

II. RELATED WORKS

EMG-based gesture recognition has been extensively studied using both traditional machine learning and deep learning approaches. Early research primarily focused on manual feature extraction and classical machine learning models. Time-domain, frequency-domain, and statistical features such as root mean square (RMS), mean absolute value (MAV), and power spectral density (PSD) have been widely employed [9]. Machine learning algorithms such as SVM, random forests, and KNN are commonly used to classify these handcrafted features [10]. While these methods provide interpretability and require smaller datasets, they heavily depend on domain expertise for feature design and often struggle to capture the complex and non-linear relationships inherent in EMG signals [11].

Recent advances in sEMG-based gesture recognition have explored transformer architectures to address spatiotemporal modeling challenges. Dere et al. [12] demonstrated the potential of vision transformers with blind source separation preprocessing, achieving high accuracy by integrating raw HD-sEMG signals, though their framework remained sensitive to noise. Montazerin et al. [13] proposed a compact transformer that jointly models macroscopic sEMG images and microscopic motor unit dynamics, emphasizing low-latency recognition across varying electrode configurations. Zabihi et al. [14] introduced a hybrid transformer with parallel feature extraction paths, improving accuracy by fusing diverse representations. While these works leverage transformers to reduce manual feature engineering and enhance spatiotemporal learning, they primarily focus on static spatial relationships between EMG channels and overlook the dynamic nature of muscle synergies during gesture execution.

To address the limitations of static feature modeling, recent works have explored graph-based methods for EMG signal analysis. Graph convolutional networks (GCNs) enable the representation of EMG channels as nodes in a graph, with

edges encoding spatial relationships. These methods have shown promise in capturing the functional connectivity between muscles [15]. However, many existing GCN-based approaches utilize predefined or static graph structures, which fail to reflect the dynamic nature of muscle coordination during gesture execution. Additionally, these models often lack the capacity to effectively model long-range temporal dependencies in EMG signals.

Transformer architectures have emerged as powerful tools for sequence modeling in various domains, including natural language processing and bioinformatics [16]. In the context of EMG signal classification, Transformers offer the potential to attend to salient features over long time horizons. However, their application has been limited, with existing approaches focusing primarily on temporal modeling while overlooking the spatial dependencies between EMG channels. Integrating Transformers with spatial feature extraction mechanisms remains an open challenge.

Our work addresses the challenge of integrating spatial and temporal modeling for EMG signal recognition by combining the strengths of dynamic GCNs and Transformers. By introducing a dynamic graph mechanism, our approach adapts to the changing interactions between EMG channels, reflecting the physiological characteristics of muscle coordination during gesture execution. The dynamic GCN nodes are designed to capture the synergistic interplay among muscles, ensuring that the model aligns with the underlying biomechanics of hand gestures. Coupled with the temporal modeling power of Transformers, this framework effectively represents both the spatial and temporal features inherent in EMG signals while adhering to their physiological basis. This approach offers a biologically inspired solution for accurately interpreting complex EMG data.

III. METHOD

This section presents the methodology employed for EMG-based gesture recognition using the proposed Dynamic Graph Transformer-Convolutional Network (GTCN). The framework is designed to leverage both the spatial interactions and temporal dependencies of EMG signals, ensuring that the model adheres to the physiological characteristics of muscle coordination during gesture execution. The approach emphasizes dynamic graph construction to capture the evolving relationships between EMG channels and utilizes Transformer layers for long-range temporal modeling. The following subsections describe the publicly available dataset used for training and evaluation, followed by a detailed explanation of the GTCN model architecture, shown in Fig. 1.

A. Dataset

This study utilized a publicly available EMG dataset, the hand Gesture Recognition and Biometrics electroMyogram Dataset (GRABMyo) [17]. The dataset includes recordings from 43 healthy participants (mean age: 26.4 ± 2.89 years; 23 males and 20 females). EMG data were collected using monopolar electrodes placed around the forearm and wrist.

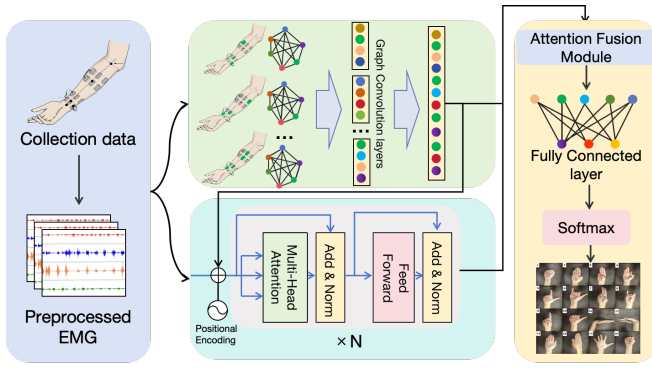


Fig. 1. GTCN model structure.

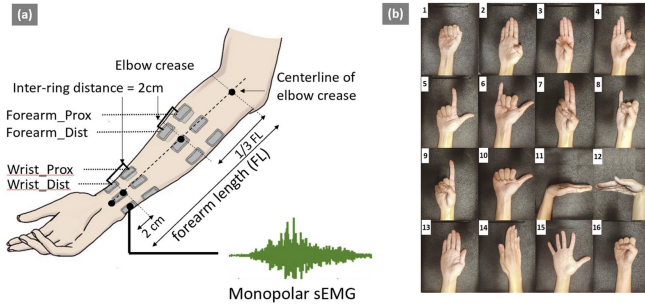


Fig. 2. EMG electrode positions and gesture categories. (a) The collection positions of eight EMG electrodes on the upper arm. (b) The execution shapes of 16 gesture categories. [17]

Specifically, 8 electrodes were positioned evenly on the forearm, and 6 electrodes were placed around the wrist at equal intervals. The data acquisition process spanned three sessions conducted on non-consecutive days (Day 1, Day 8, and Day 29), ensuring variability and robustness in the recordings. During each session, participants performed 16 static hand and wrist gestures, repeated seven times, followed by a rest trial. This protocol was designed to capture a diverse range of muscle activation patterns associated with gesture execution. The GRABMyo dataset is notable for its scale, comprising 129 recording sessions (43 participants $\times$ 3 days), making it one of the most comprehensive EMG datasets in terms of session count. This extensive dataset provides a robust foundation for training and evaluating advanced models like the proposed GTCN.

B. Graph Transformer-Convolutional Network

The proposed GTCN is designed to effectively model the spatial-temporal relationships in multi-channel EMG signals, leveraging the physiological characteristics of muscle coordination during hand and wrist gestures. This section provides a detailed description of the GTCN framework, its underlying principles, and the data flow through the model.

The GTCN integrates dynamic GCN and Transformer to effectively capture the spatial-temporal relationships in multi-channel EMG signals. This model is designed to leverage the physiological coordination between muscles during hand gestures, making it well-suited for hand gesture recognition from preprocessed EMG data.

1) *EMG Data Definition:* In this study, the input EMG data consists of multi-channel electromyographic signals. The EMG signals are recorded by C channels (i.e., muscle sensors) over T time steps, where each time step t contains a $C \times 1$ vector. The format of the EMG data can be represented as:

$$\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T], \quad \mathbf{x}_t \in \mathbb{R}^C \quad (1)$$

where $\mathbf{X}$ represents the raw EMG signal sequence, T is the number of time steps, C is the number of channels, and $\mathbf{x}_t$ is the C -dimensional EMG signal vector at the t -th time step.

2) *Dynamic GCN Feature Extraction:* In this algorithm, the first step is to model the EMG data as a graph. Each muscle sensor corresponds to a node in the graph, and the connections between nodes are determined by the correlation between the recorded signals. Specifically, the Pearson correlation coefficient is used to measure the similarity between different EMG recording points (channels), which is calculated based on each sample. For any pair of channels i and j , the connection weight w_{ij} is calculated as the Pearson correlation coefficient:

$$w_{ij} = \frac{\text{cov}(\mathbf{x}_i, \mathbf{x}_j)}{\sigma_{\mathbf{x}_i} \sigma_{\mathbf{x}_j}} \quad (2)$$

where $\text{cov}(\mathbf{x}_i, \mathbf{x}_j)$ denotes the covariance between the signals from channels i and j , and $\sigma_{\mathbf{x}_i}$ and $\sigma_{\mathbf{x}_j}$ are the standard deviations of the signals from channels i and j , respectively.

This weight matrix is used to construct the graph adjacency matrix $\mathbf{A}$, and graph convolution operations are performed to extract dynamic graph features. Given that EMG signals vary over time, the graph structure should also change dynamically. Specifically, the graph structure at each time step t is denoted by $\mathbf{A}_t$, which depends on the correlation between the channels at the current time step.

The graph features extracted by GCN are then combined with the raw EMG data and fed into a Transformer model for temporal sequence modeling. To encode temporal information, position encoding is applied to the graph features. Assuming the GCN output for time step t is $\mathbf{G}_t \in \mathbb{R}^{C \times F}$, where F represents the feature dimension of each node, the position encoding is computed as follows:

$$\mathbf{PE}_t[i] = \sin\left(\frac{t}{10000^{2i/F}}\right) \quad \text{if } i \text{ is even} \quad (3)$$

$$\mathbf{PE}_t[i] = \cos\left(\frac{t}{10000^{2i/F}}\right) \quad \text{if } i \text{ is odd} \quad (4)$$

Then, the raw EMG data, position encoding $\mathbf{PE}_t$, and graph features $\mathbf{G}_t$ are concatenated to form the input to the Transformer:

$$\mathbf{H}_t = \text{concat}(\mathbf{X}_t, \mathbf{PE}_t, \mathbf{G}_t) \quad (5)$$

where $\mathbf{H}_t$ is the fused feature vector containing the raw EMG data, position encoding, and graph features, which serves as the input to the Transformer model.

3) *Transformer Module*: In the Transformer model, which consists of 4 stacked Transformer layers, a multi-head self-attention mechanism is employed to capture temporal dependencies and long-range interactions in the EMG signal sequence. For each time step t , the input is processed using multiple self-attention heads ($=8$), which aggregate information from different parts of the sequence, allowing the model to identify important time periods for gesture recognition. The multi-head self-attention mechanism can be defined as:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right) \mathbf{V} \quad (6)$$

where $\mathbf{Q}, \mathbf{K}, \mathbf{V}$ represent the query, key, and value matrices, respectively, and d_k is the dimensionality of the keys.

By using multi-head attention, the model can capture different aspects of the temporal data, which is crucial for identifying the key time segments that contribute to the gesture classification.

4) *Feature Fusion and Classification*: The temporal features extracted by the Transformer and the graph features extracted by GCN are fused in this step. The feature fusion strategy employed here is the Attentional Fusion Model (AFM), which uses an attention mechanism to weight the contribution of different feature sources. Specifically, if the Transformer output at time step t is $\mathbf{T}_t \in \mathbb{R}^F$ and the GCN output is $\mathbf{G}_t \in \mathbb{R}^C$, their fused representation is computed as:

$$\mathbf{F}_t = \text{AFM}(\mathbf{T}_t, \mathbf{G}_t) \quad (7)$$

The AFM mechanism computes the similarity between the input features and dynamically assigns weights to each source feature based on their relevance to the final decision.

Finally, the fused features $\mathbf{F}_t$ are passed through a fully connected layer for classification:

$$\hat{y}_t = \text{softmax}(\mathbf{W}\mathbf{F}_t + \mathbf{b}) \quad (8)$$

where $\hat{y}_t$ is the predicted probability distribution of the gesture classes, and $\mathbf{W}$ and $\mathbf{b}$ are the weights and biases of the fully connected layer, respectively.

To improve the generalization ability of the model, dropout regularization with a rate of 0.2 is applied in both the fully connected layers and the self-attention mechanism of the Transformer. The dropout operation randomly drops a proportion of neurons during training to prevent overfitting. The dropout operation can be expressed as:

$$\mathbf{y}_{\text{dropout}} = \text{Dropout}(\mathbf{y}, p) \quad (9)$$

where $\mathbf{y}$ is the input feature, p is the probability of dropping a neuron, and $\mathbf{y}_{\text{dropout}}$ is the output after dropout regularization.

5) *Loss Function*: For the multi-class gesture classification task, the loss function commonly used is the cross-entropy loss, which is defined as:

$$\mathcal{L} = - \sum_{c=1}^C y_c \log(\hat{y}_c) \quad (10)$$

where y_c represents the true probability distribution of the labels, $\hat{y}_c$ represents the predicted probability distribution, and C is the number of gesture classes.

This algorithm leverages a dynamic graph structure to model the spatial relationships between muscle sensors, followed by GCN operations to extract dynamic graph features. These features are then combined with the raw EMG data and position encoding and fed into a Transformer model for temporal sequence modeling. The multi-head self-attention mechanism captures the temporal dependencies and highlights the important time segments for gesture recognition. The Attentional Fusion Model (AFM) is used to fuse the graph and temporal features, which are then classified using a fully connected layer. Dropout regularization enhances the model's generalization ability. This framework is well-suited for multi-channel EMG gesture recognition tasks, effectively combining spatial and temporal information for accurate classification.

IV. RESULTS AND DISCUSSIONS

In this section, we present the results of five-fold cross-validation experiments using the GRABMyo dataset. The performance of the GTCN model was evaluated based on the gesture classification accuracy, and comparisons were made against traditional machine learning methods, such as SVM and Random Forests (RF), as well as deep learning approaches, including CNN and RNN [18].

Fig. 3 shows the accuracy of various machine learning and deep learning models for EMG gesture recognition. Our proposed GTCN algorithm stands out with the highest accuracy of 97.24%. This represents a significant improvement in accuracy compared to traditional machine learning models: LDA has an accuracy of 85.37%, SVM has an accuracy of 91.46%, KNN has an accuracy of 91.79%, Random Forest has an accuracy of 91.83%, CNN has an accuracy of 94.68%, and the CNN-LSTM model has an accuracy of 94.71%.

The GTCN algorithm achieves the highest accuracy of 97.24% in the EMG gesture recognition task. This is mainly due to its architecture, which combines graph convolutional networks (GCNs) and Transformer structures, enabling the model to fully capture the spatiotemporal characteristics of EMG signals. First, GTCN uses dynamic graph modeling, treating EMG acquisition points as nodes in a graph and employing the Pearson correlation coefficient to measure relationships between different channels, thereby constructing a dynamically weighted adjacency matrix. This allows the model to adaptively adjust connection weights between EMG acquisition points, avoiding the static assumptions about muscle relationships in traditional methods and improving the representation of spatial features. Secondly, GCN extracts features from the graph structure, effectively capturing muscle association patterns, which enhances the model's ability to learn spatial dependencies in gesture recognition. Traditional machine learning methods rely on manual feature extraction, making it difficult to characterize the complex relationships in EMG data. CNNs, on the other hand, can only learn spatial information through local convolution

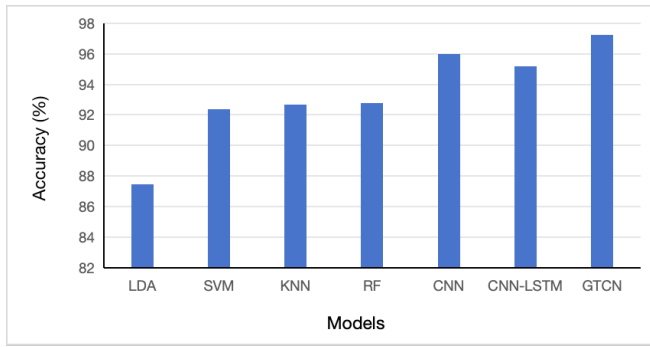


Fig. 3. Gesture recognition using traditional machine learning and deep learning models.

operations and cannot establish global dependencies between EMG acquisition points. This gives GTCN a significant advantage in spatial feature extraction. In terms of temporal feature modeling, GTCN uses a Transformer for time-series analysis. Compared with CNN and CNN-LSTM architectures, the Transformer's multi-head self-attention mechanism dynamically allocates weights, allowing the model to adaptively focus on key gesture-related signals in different time segments, thereby enhancing temporal feature learning. Compared with traditional methods, GTCN not only models the spatiotemporal dependencies of EMG data simultaneously but also dynamically adjusts temporal weights and muscle connections, making the model more robust across different gesture categories. This series of design choices ultimately enables GTCN to achieve a maximum accuracy of 97.24%, significantly better than deep learning methods such as CNN (94.68%) and CNN-LSTM (94.71%), proving its effectiveness in EMG gesture recognition tasks.

V. CONCLUSIONS

In this study, we proposed a GTCN model for hand gesture recognition using preprocessed EMG data. By leveraging the dynamic nature of the graph structure and the ability of the model to simultaneously capture both temporal and spatial dependencies, the GTCN demonstrated superior performance in classifying hand gestures compared to traditional machine learning and other deep learning approaches. The model achieved an accuracy of 97.24% on the GRABMyo dataset, which highlights its effectiveness in handling the complex, high-dimensional nature of EMG signals.

The dynamic graph convolution mechanism was key in modeling the interdependencies between different EMG channels, reflecting the physiological coordination of muscle activity during hand gestures. This novel approach allows the model to adaptively learn the spatial relationships between channels, enhancing its generalization ability and robustness to variations in signal quality and noise.

Future work will focus on expanding the model to handle more complex hand gestures, incorporating real-time performance considerations, and exploring the use of multi-modal data (such as integrating EMG with EEG sensors) to further improve recognition accuracy and robustness. Additionally,

we plan to investigate the potential of transfer learning for leveraging pre-trained models in cross-subject and cross-dataset scenarios, which could significantly enhance the model's applicability in real-world applications.

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